

Stuart Clark

ML Engineer | Explainable AI | Risk & Compliance

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PROFESSIONAL SUMMARY

ML Engineer with a First Class MSc in Computer Science & AI (Northumbria University, 2026) and 18 years of senior-level experience in occupational health and safety. Dissertation research centred on fine-tuning DistilBERT and ModernBERT transformer models on 180,000+ OSHA incident records to predict a 5-level workplace severity classification, with SHAP explainability integrated to produce auditable, regulation-mapped outputs defensible to regulators and non-technical stakeholders. Three production projects deployed across Railway and AWS ECS Fargate demonstrating a full ML engineering lifecycle from data ingestion to structured output. Targeting roles in risk-tech, RegTech, insurtech, and AI governance where domain expertise in regulated, high-stakes environments is a genuine differentiator.

TECHNICAL SKILLS

ML / AI	PyTorch, TensorFlow, Scikit-learn, DistilBERT, ModernBERT, Transformers, NLP, SHAP (Explainable AI), Optuna
LLMs & GenAI	Gemini 2.5 Flash, RAG (Sequential), Prompt Engineering, Vector Search
Vector & Search	ChromaDB, Sentence Transformers, HuggingFace
APIs & Backend	FastAPI, Pydantic v2, Streamlit, REST APIs
Cloud & Deploy	AWS ECS Fargate, AWS S3, Railway, Vercel, Docker
Doc Processing	PyMuPDF, Tesseract OCR, Multimodal AI
ML Tooling	MLflow, Experiment Tracking, Structured Logging
Languages	Python (primary), Java
Domain	Occupational Health & Safety, Risk Assessment, RIDDOR, NEBOSH, Regulatory Compliance, Incident Investigation, Local Government, Food Safety

KEY PROJECTS & RESEARCH

MSc Dissertation: NLP Severity Classifier for Workplace Incidents · Northumbria University, 2025–26

- Fine-tuned DistilBERT and ModernBERT transformer models (PyTorch) on 180,000+ OSHA incident records to predict a 5-level severity classification, grounded in UK RIDDOR legislation and time-loss economic data; used Optuna for hyperparameter optimisation and strategic class weighting to address severe data imbalance.
- Integrated SHAP (Explainable AI) to surface token-level decision reasoning — enabling auditable outputs readable by non-technical H&S managers and enforcement officers, directly addressing the ML black-box problem in regulated domains.
- Key finding: models excelled at severity extremes (High / No Harm); SHAP revealed a systematic caution bias (predicting Major over Minor) — the correct failure mode for safety-critical deployment, demonstrating that XAI is essential for responsible AI in regulated environments.

Care Home Incident Triage API · Deployed — AWS ECS Fargate

- REST API accepting health and social care workplace incident narratives and returning a structured triage analysis via a sequential RAG pipeline — fine-tuned DistilBERT severity classifier combined with ChromaDB vector retrieval over RIDDOR 2013, HSG220, and OSHA knowledge bases.
- Gemini 2.5 Flash LLM performs regulatory reasoning and causal factor extraction; outputs include severity prediction with RIDDOR mapping, causal analysis, mitigation guidance, and population-level pattern signals from OSHA incident data.
- Stack: Python, FastAPI, Pydantic v2, DistilBERT, ChromaDB, Sentence Transformers, Gemini 2.5 Flash, AWS ECS Fargate, Vercel.

OHS Document Intelligence Pipeline · *Live deployment — Railway*

- End-to-end pipeline automating structured data extraction from UK safety compliance certificates (LOLER thorough examination, PSSR pressure vessels) across PDF, JPG, and PNG formats including scanned documents.
- Pipeline: PyMuPDF (digital PDFs) / Tesseract OCR (scanned inputs) → document classification → Gemini 2.5 Flash multimodal field extraction → Pydantic v2 validated JSON with generated alerts and calendar reminders; Streamlit interface for stakeholder demonstration.
- Stack: Python, FastAPI, Pydantic v2, Gemini 2.5 Flash, PyMuPDF, Tesseract OCR, Railway, Streamlit.

PROFESSIONAL EXPERIENCE

MSc Student — Computer Science & AI | Northumbria University (Distance Learning) *Oct 2023 – Jan 2026*

- Full-time study completed with a First Class grade. Modules: Machine Learning, Artificial Intelligence, NLP, Software Engineering, Python and Java programming. See Projects section for dissertation and deployed project detail.

Career Break — Full-time Parenting

Sep 2022 – Oct 2023

- Relocated to Kathmandu, Nepal. Completed preparatory Computer Science coursework ahead of MSc enrolment in October 2023.

Safety Consultant | Self-Employed

Mar 2022 – Sep 2022

- Independent site inspections, compliance audits, and risk assessments for food business operators; detailed written findings reports for client managers, operations teams, and environmental health enforcement officers.

Sabbatical — Professional Development

Mar 2019 – Mar 2022

- Completed the NEBOSH National Diploma (Level 6) — a degree-level qualification in occupational health and safety management.

Senior Health and Safety Consultant | NAVITAS / ESB

May 2011 – Mar 2019

- Delivered H&S and food safety compliance services across a major client portfolio; dedicated client manager for Tesco Hospitality — coordinating compliance programmes and reporting outcomes to senior stakeholders.
- Conducted physical site inspections and compliance audits, producing detailed findings reports for operations managers, H&S managers, and regulatory enforcement officers.
- Investigated workplace incidents and food safety complaints — gathering evidence, identifying root causes, and communicating findings across technical and non-technical audiences.
- Delivered accredited Level 2 and Level 3 OHS and food safety training, translating complex regulatory requirements into clear, accessible content for frontline staff.

Local Government Regulatory Officer | North Devon DC / Mid Devon DC / Bampton Consultancy *Sep 2003 – Apr 2011*

- Early career roles across local authority environmental health and private sector housing regulation — statutory enforcement, evidence gathering, structured document assessment, and legal compliance across multiple client sites.

EDUCATION & CERTIFICATIONS

MSc Computer Science & Artificial Intelligence · First Class

2023 – 2026

Northumbria University, UK (Distance Learning)

BSc (Hons) Environmental Health

1999 – 2003

Cardiff Metropolitan University, UK · Graduate of EHRB · MCIEH

Professional Certifications

- DipNEBOSH — National Diploma in Occupational Health and Safety (Level 6)
- MCIEH — Member of the Chartered Institute of Environmental Health